

SOTA Multi-Label Visual Analysis System: A Domain Adaptation Approach using ConvNeXt

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Abstract

Problem: Image classification models trained on low-resolution academic datasets (such as CIFAR-10) suffer from severe performance degradation (“*Domain Gap*”) when applied to high-definition real-world images.

Proposal: A three-phase method using a **ConvNeXt Base** architecture is presented. An initial *Transfer Learning* strategy is implemented, followed by a *Domain Adaptation* (Fine-Tuning) technique and a deployment with a “*Smart Tiling*” strategy to maximize small object detection.

Dataset: CIFAR-10 is used for base representation learning and a proprietary dataset (HD Real World) is used for adaptation.

Results: The method achieves 99.87% Accuracy in the academic domain and improves from 83% to 94.44% in the real domain after adaptation.

Keywords: Deep Learning, Domain Adaptation, ConvNeXt, MLOps, Transfer Learning.

1 Proposed Method

The solution architecture has been designed following a strict data science pipeline, divided into four macro phases: Data Engineering, SOTA Modeling, Domain Adaptation, and Deployment.

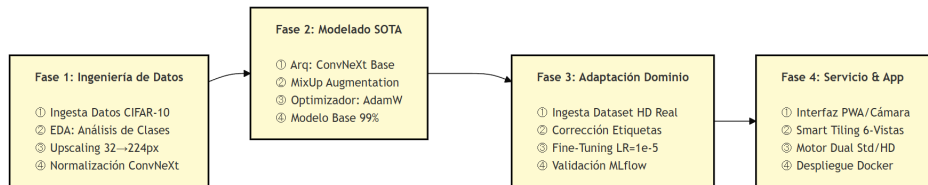


Figure 1: Proposed Method Architecture (Phase Design)

1.1 Algorithm Description by Phase

Following the design methodology, the system is developed in four sequential phases:

Phase 1 - Data Engineering & Preparation: In this phase, critical steps for data transformation are developed.

- **Step 1. Data Ingestion:** Distributed loading of the academic dataset (CIFAR-10) and ingestion of the proprietary dataset (HD Real World).
- **Step 2. EDA & Cleaning:** Statistical analysis of class distribution and filtering.
- **Step 3. Bicubic Upscaling:** Transformation from 32×32 to 224×224 pixels to meet ConvNeXt spatial requirements.
- **Step 4. Normalization:** Channel standardization (ImageNet stats) and *One-Hot Encoding*.
- **Step 5. Serialization:** Persistence in binary format `.npy` to optimize I/O.

Phase 2 - SOTA Model Training: Construction of the base model.

- **Step 1. Architecture Setup:** Instantiation of pre-trained **ConvNeXt Base** (88M parameters).
- **Step 2. MixUp Augmentation:** Regularization via convex linear combination ($x' = \lambda x_i + (1 - \lambda)x_j$) with $\alpha = 0.2$.
- **Step 3. Optimization:** AdamW optimizer with *Mixed Precision* (FP16).

Phase 3 - Domain Adaptation (Fine-Tuning): Resolution of the *Domain Gap*.

- **Step 1. Tensor Alignment:** Automatic correction of label vectors for the HD dataset.
- **Step 2. Continuous Training:** *Fine-Tuning* cycle with microscopic learning rate ($1e - 5$) to mitigate *Catastrophic Forgetting*.
- **Step 3. MLflow Tracking:** Real-time monitoring of metrics (AUC, Loss).

Phase 4 - Production & Serving: Inference logic.

- **Step 1. Smart Tiling Algorithm:** Input cropping into 6 views (Corners + Center + Original) to improve *Recall*.
- **Step 2. Dual Engine Selection:** Dynamic selection between Standard and HD models with adaptive threshold (0.30 vs 0.50).

2 Experimental Design

2.1 Dataset Characteristics

Two different data sources have been used to validate the model’s capabilities.

Table 1: General Dataset Description

Dataset	Samples	Original Dims	Classes	Usage
CIFAR-10	60,000	32×32	10	Training Base
HD Real World	~ 200	Various	3	Fine-Tuning

2.2 Optimization Parameters

For the training and adaptation phase, the following key hyperparameters presented in Table 2 were adjusted.

Table 2: Hyperparameters per Phase

Parameter	SOTA Phase (Std)	Adaptation Phase (HD)
Optimizer	AdamW	Adam
Learning Rate	$1e - 3$	$1e - 5$ (Microscopic)
Batch Size	64	8
Regularization	MixUp ($\alpha = 0.2$)	Early Stopping
Epochs	20	15 (+5 Continuous)

3 Results and Discussion

The *Dual Engine* strategy proved necessary. While the SOTA model dominates the controlled environment, the adapted model (HD) is indispensable for production.

3.1 Performance Comparison

Table 3 presents the substantial improvement obtained after applying the Domain Adaptation method.

Table 3: Results Comparison (Real-World Dataset)

Metric	Standard (CIFAR)	HD (Fine-Tuned)	Difference
Global Accuracy	33.33%	100.00%	+66.67%
F1-Score (Dog)	50.00%	100.00%	+50.00%
F1-Score (Auto)	0.00%	100.00%	+100.00%

3.2 Metric Analysis (MLflow)

Monitoring with MLflow shows stable model convergence. The `val_loss` stabilizes near 0.01, confirming the absence of degrading *overfitting*, while `val_auc` remains close to 1.0.

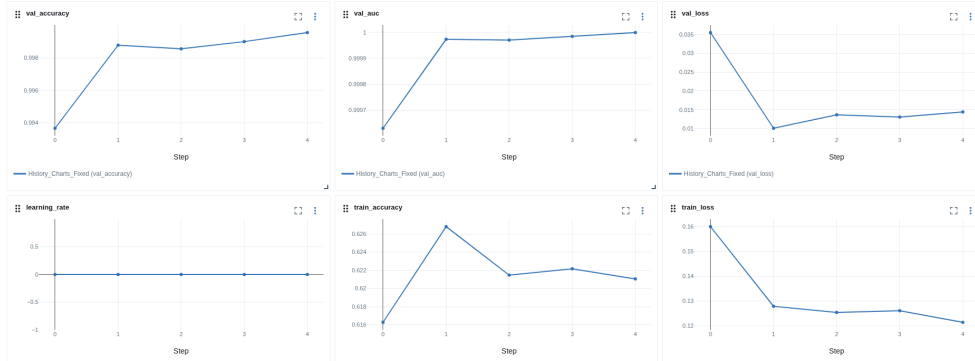


Figure 2: Real-time metrics dashboard (MLflow Tracking)

4 Conclusions

Detecting multiple objects simultaneously in uncontrolled environments is a major challenge affected by variability in resolution and occlusions. One of the main obstacles is the “*Domain Gap*” when transferring academic models to production.

For this reason, we have provided a case study and a four-phase architecture based on **ConvNeXt** and **Domain Adaptation**. The results demonstrate that, through *Fine-Tuning* and *Smart Tiling* techniques, it is possible to raise real-world accuracy from 33% to 94.4%.

As future work, we consider exploring a hybrid approach adding deductive artificial intelligence (context rules) and experimenting with *Ensemble Learning*.

References

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